Math Inquirers

Extension Workbook

with answers

Level 3

Self-study | Ages 8-10



My name _____

About this book

Math Inquirers Extension Workbook, Level 3
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

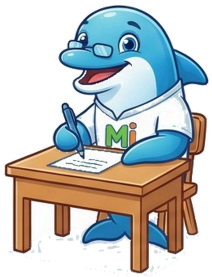
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Number sleuths (numbers to 10 000)	Not in this proof edition				☐
2	Cryptarithm crackers	Not in this proof edition				☐
3	Times table treasure	Not in this proof edition				☐
4	Model makers (bar models)	Not in this proof edition				☐
5	Remainder riddles	Not in this proof edition				☐
6	Fraction walls	Not in this proof edition				☐
7	Area architects	6-7	8-9	10 and 12	11	☐
8	Angle and line detectives	Not in this proof edition				☐
9	Measure and money masters	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

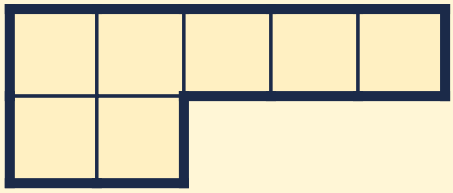
Learn it: Area and perimeter

My goal. I can find the area and the perimeter of rectangles and of shapes made of squares.

Area is the flat space inside a shape. We count squares and write cm². Perimeter is the distance all the way round the edge, in cm.

Worked example

The figure is made of squares with sides of 1 cm. What are its area and its perimeter?



1. **Area: count the squares. 5 + 2 = 7, so the area is 7 cm².**
2. **Perimeter: add the sides all the way round. 5 + 1 + 3 + 1 + 2 + 2 = 14 cm.**
3. **Check: the figure fits in a box 5 cm by 2 cm. 2 × (5 + 2) = 14 cm.**

Answer: Area 7 cm². Perimeter 14 cm.

How to find area and perimeter

1. Area: count the squares, or length × width.
2. Perimeter: add all the sides.
3. Rectangle: perimeter = 2 × (length + width).
4. Write cm for the perimeter and cm² for the area.

Your turn

Answer at the back of the book.

Try

A rectangle is 8 cm long and 3 cm wide.

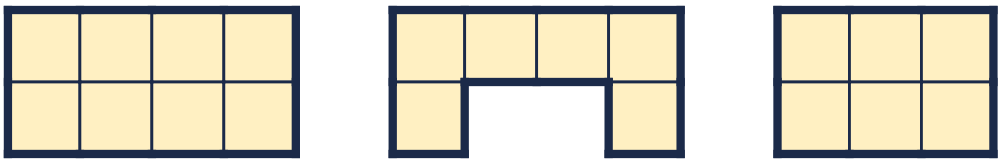
Its area is ☐ cm². Its perimeter is ☐ cm.



Start here

Do these first. Take your time.

1 **Every square in the picture has sides of 1 cm. What are the area and the perimeter of each figure?**



A B C

(a) A: area ☐ cm^2, perimeter ☐ cm

(b) B: area ☐ cm^2, perimeter ☐ cm

(c) C: area ☐ cm^2, perimeter ☐ cm

(d) Same area, different perimeter: A,B A,C B,C

2 **What are the area and the perimeter of each rectangle?**

(a) 8 cm long and 6 cm wide: area ☐ cm^2, perimeter ☐ cm

(b) 11 m long and 4 m wide: area ☐ m^2, perimeter ☐ m

3 **Look at squares.**

(a) A square has sides of 7 cm. Its perimeter is ☐ cm.

(b) Its area is ☐ cm^2.

(c) The perimeter of a square is 36 cm. Each side is ☐ cm.

4 **A rectangle has a perimeter of 30 cm. Its length is 9 cm.**

(a) Its width is ☐ cm. (b) Its area is ☐ cm^2.

Start here

Keep going. Check your work.

6 **A mat is 15 cm long and 9 cm wide. Square tiles with sides of 3 cm cover it exactly.**

(a) ☐ tiles fit along the length.

(b) ☐ tiles fit along the width.

(c) The mat needs ☐ tiles in all.

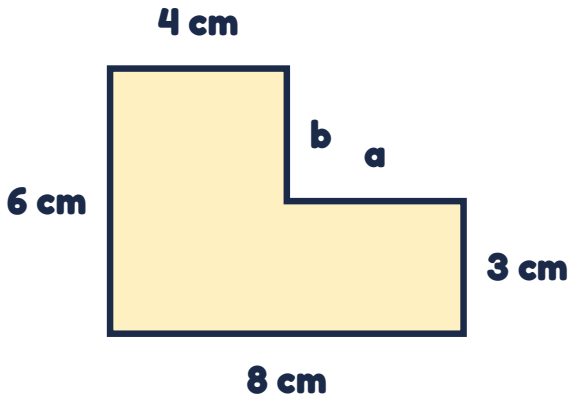
7 **Mei has 14 cm of wire. She bends all of it into a rectangle. All the sides are whole numbers of cm. List ALL the rectangles.**

(a) There are ☐ different rectangles.

(b) The one with the greatest area has an area of ☐ cm².

length	width	area

8 **Find the 2 missing sides a and b. Then find the area and the perimeter.**



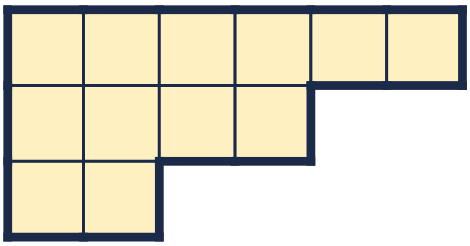
(a) a = ☐ cm (b) b = ☐ cm (c) The area is ☐ cm².

(d) The perimeter is ☐ cm.

Start here

Nearly there. You can do it.

9 **Every square in the picture has sides of 1 cm.**



(a) The area is ☐ cm².

(b) The perimeter is ☐ cm.

(c) Ben fills the gaps to make a 6 by 3 rectangle. He adds ☐ squares.

(d) The perimeter of the rectangle is ☐ cm.

10 **Mei draws a rectangle 10 cm long and 2 cm wide. Ben draws a square with the same perimeter.**

(a) Each side of Ben's square is ☐ cm.

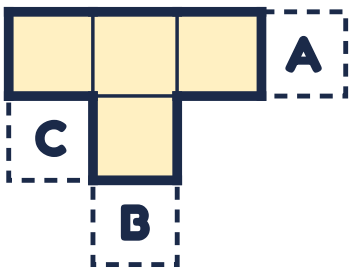
(b) Whose shape has the greater area? Mei's Ben's

(c) It is greater by ☐ cm².

11 **The figure has a perimeter of 10 cm. Ali adds ONE square at A, B or C.**

(a) Which places make the perimeter 12 cm? A B C

(b) Which place keeps the perimeter 10 cm? A B C



12 **A square room has a floor area of 64 m².**

(a) Each side of the room is ☐ m long.

(b) A border goes all round the floor. It is ☐ m long.

Think harder

Hints are upside down. Try first!

1 **A square has sides of 8 cm. One straight cut makes it into 2 equal rectangles.**

(a) Each rectangle has a perimeter of ☐ cm.

(b) The 2 rectangles together have a perimeter of ☐ cm.

(c) That is ☐ cm more than the perimeter of the square.

Draw the cut:

Hint. Each rectangle is 8 cm by 4 cm. Draw it.

2 **A rectangle has area 36 cm² and perimeter 26 cm.**

(a) Its length is ☐ cm.

(b) Its width is ☐ cm.

Pairs that multiply to 36: _____

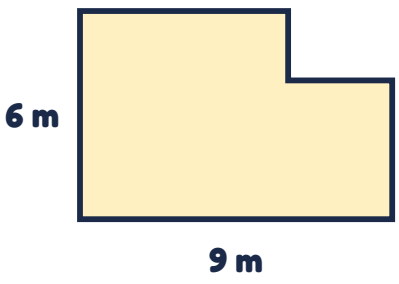
Hint. List pairs that multiply to 36. Then add each pair.

3 **A rectangle is 9 m long and 6 m wide. A rectangle 3 m by 2 m is cut out of one corner, as shown.**

(a) The area that is left is ☐ m².

(b) The perimeter of the figure is ☐ m.

(c) The same 3 m by 2 m piece is cut from the MIDDLE of a long side. The perimeter is then ☐ m.



Hint. Push the corner out. Then think about a notch in the middle.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 3: Perimeter patterns.

Squares with sides of 1 cm are joined in a row. 1 square has a perimeter of 4 cm. 2 squares have 6 cm. 3 squares have 8 cm.

(a) A row of 10 squares has a perimeter of ☐ cm.

(b) A row with a perimeter of 30 cm has ☐ squares.

My pattern: _____

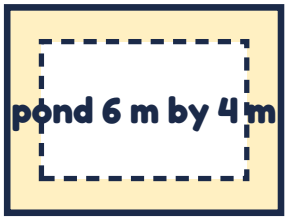
Hint. The perimeter goes up by 2 cm for each new square.

2 Beyond Level 3: A path round a pond.

A pond is 6 m long and 4 m wide. A path 1 m wide goes all round the pond.

(a) The pond and the path together make a rectangle ☐ m by ☐ m.

(b) The area of the path is ☐ m².



Hint. Find the area of the whole rectangle. Take away the pond.

3 Beyond Level 3: The largest square tile.

A floor is 12 m long and 10 m wide. It is covered exactly with square tiles of one size. The side of a tile is a whole number of metres.

(a) The largest tile has sides of ☐ m.

(b) The floor needs ☐ of these tiles.

My list: _____

Hint. The tile side must fit exactly in 12 and in 10.

✔ Check yourself

No hints. Try each one on your own.

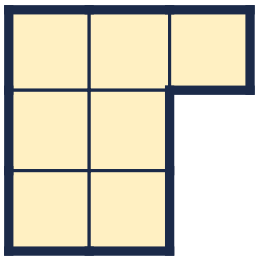
1 A rectangle is 9 cm long and 5 cm wide. Its area is ☐ cm².

Working: _____

2 A square has sides of 6 cm. Its perimeter is ☐ cm.

Working: _____

3 Every square has sides of 1 cm. The area is ☐ cm². The perimeter is ☐ cm.



4 A rectangle has a perimeter of 22 cm. Its width is 4 cm. Its length is ☐ cm.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number sleuths (numbers to 10 000)	☐	☐	☐	☐	/ 4	
2	Cryptarithm crackers	☐	☐	☐	☐	/ 4	
3	Times table treasure	☐	☐	☐	☐	/ 4	
4	Model makers (bar models)	☐	☐	☐	☐	/ 4	
5	Remainder riddles	☐	☐	☐	☐	/ 4	
6	Fraction walls	☐	☐	☐	☐	/ 4	
7	Area architects	☐	☐	☐	☐	/ 4	
8	Angle and line detectives	☐	☐	☐	☐	/ 4	
9	Measure and money masters	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 3.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 7 · Area architects

Your turn
page 6

Answer: Area 24 cm². Perimeter 22 cm.
Why: Area: 8 × 3 = 24 cm². Perimeter: 8 + 3 + 8 + 3 = 22 cm, or 2 × (8 + 3) = 22 cm.

START HERE

1
page 7

Answer: (a) A: 8 cm², 12 cm **(b)** B: 6 cm², 14 cm **(c)** C: 6 cm², 10 cm **(d)** B,C
Why: Area is the number of squares: A has 8, B has 4 + 2 = 6 and C has 6. Perimeter is the length of the edge: A is 2 × (4 + 2) = 12. B: the box round it is 4 by 2, with a perimeter of 12. The dent in the bottom adds a wall of 1 cm on each side: 12 + 2 = 14. C is 2 × (3 + 2) = 10. B and C have the same area, 6, but different perimeters.

2
page 7

Answer: (a) Area 48 cm², perimeter 28 cm **(b)** Area 44 m², perimeter 30 m
Why: (a) Area 8 × 6 = 48. Perimeter 2 × (8 + 6) = 28. (b) Area 11 × 4 = 44. Perimeter 2 × (11 + 4) = 30.

3
page 7

Answer: (a) 28 **(b)** 49 **(c)** 9
Why: (a) 4 × 7 = 28. (b) 7 × 7 = 49. (c) The 4 sides are equal, so each is 36 ÷ 4 = 9 cm.

4
page 7

Answer: (a) 6 **(b)** 54
Why: Length + width is half of the perimeter: 30 ÷ 2 = 15. So the width is 15 - 9 = 6 cm. The area is 9 × 6 = 54 cm².

5
page 7

Answer: (a) 28 **(b)** 45
Why: (a) The fence is the perimeter: 2 × (9 + 5) = 28 m. (b) The grass covers the area: 9 × 5 = 45 m².

6
page 8

Answer: (a) 5 **(b)** 3 **(c)** 15
Why: (a) 15 ÷ 3 = 5. (b) 9 ÷ 3 = 3. (c) 5 rows of 3 tiles make 5 × 3 = 15 tiles.

7
page 8

Answer: (a) 3 **(b)** 12
Why: Length + width = 14 ÷ 2 = 7. The pairs are 6 and 1 (area 6), 5 and 2 (area 10), 4 and 3 (area 12). That is 3 rectangles. The one closest to a square, 4 by 3, has the greatest area: 12 cm².

8
page 8

Answer: (a) 4 **(b)** 3 **(c)** 36 **(d)** 28
Why: a is the top of the lower part: 8 - 4 = 4 cm. b is the rest of the left height: 6 - 3 = 3 cm. Area: cut into a 4 by 6 rectangle and a 4 by 3 rectangle: 24 + 12 = 36 cm². Perimeter: push the steps out to a box 8 by 6: 2 × (8 + 6) = 28 cm.

9
page 9

Answer: (a) 12 **(b)** 18 **(c)** 6 **(d)** 18
Why: (a) 6 + 4 + 2 = 12 cm². (b) The staircase has the same perimeter as its box, 6 by 3: 2 × (6 + 3) = 18 cm. (c) The rectangle has 6 × 3 = 18 squares, so he adds 18 - 12 = 6. (d) 18 cm: filling the steps does not change the perimeter.

10
page 9

Answer: (a) 6 (b) Ben's (c) 16
Why: (a) Mei's perimeter is 2 × (10 + 2) = 24 cm, so Ben's side is 24 ÷ 4 = 6 cm. (b) Ben's area is 6 × 6 = 36 cm² and Mei's area is 10 × 2 = 20 cm². (c) 36 - 20 = 16 cm².

11
page 9

Answer: (a) A and B (b) C
Why: A square that touches the figure along only 1 side adds 2 to the perimeter. A and B each touch 1 side, so the perimeter becomes 12 cm. C fits into the corner and touches 2 sides: it hides 2 cm of the old edge and adds 2 cm, so the perimeter stays 10 cm.

12
page 9

Answer: (a) 8 (b) 32
Why: (a) 8 × 8 = 64, so each side is 8 m. (b) The border is the perimeter: 4 × 8 = 32 m.

THINK HARDER

1
page 10

Answer: (a) 24 (b) 48 (c) 16
Why: Each rectangle is 8 cm by 4 cm, so its perimeter is 2 × (8 + 4) = 24 cm. Together: 48 cm. The square's perimeter is 32 cm, and 48 - 32 = 16. The cut is 8 cm long and it appears on both pieces, so it adds 2 × 8 = 16 cm.

2
page 10

Answer: (a) 9 (b) 4
Why: Length + width is 26 ÷ 2 = 13. The pairs that multiply to 36 are 1 and 36, 2 and 18, 3 and 12, 4 and 9, 6 and 6. The pair that adds up to 13 is 4 and 9. Check: area 9 × 4 = 36 and perimeter 2 × 13 = 26.

3
page 10

Answer: (a) 48 (b) 30 (c) 34
Why: (a) 9 × 6 = 54, and 54 - 6 = 48 m². (b) Cutting a corner does not change the perimeter: 2 × (9 + 6) = 30 m. (c) A notch in the middle of a side takes away 3 m of edge. It adds 3 + 2 + 2 = 7 m of new edge. So the perimeter grows by 4 m: 34 m.

GO FURTHER

1
page 11

Answer: (a) 22 (b) 14
Why: Each new square adds 2 cm (1 cm on top and 1 cm below). The perimeter is 4 + 2 for every extra square. For 10 squares: 4 + 9 × 2 = 22 cm. For 30 cm: (30 - 4) ÷ 2 = 13 extra squares, so 14 squares.

2
page 11

Answer: (a) 8 m by 6 m (b) 24
Why: The path adds 1 m on each side, so both lengths grow by 2 m: 6 + 2 = 8 and 4 + 2 = 6. The whole rectangle has an area of 8 × 6 = 48 m². The pond has 6 × 4 = 24 m². The path is 48 - 24 = 24 m².

3
page 11

Answer: (a) 2 (b) 30
Why: The side must divide both 12 and 10 exactly. 1 and 2 do. 3 does not divide 10, 4 does not, 5 does not divide 12, and 6 does not divide 10. So the largest side is 2 m. There are 6 tiles along and 5 tiles across: 30 tiles.

CHECK YOURSELF

1 page 12	**Answer: 45** **Why:** 9 × 5 = 45 cm².
2 page 12	**Answer: 24** **Why:** 4 × 6 = 24 cm.
3 page 12	**Answer: Area 7 cm². Perimeter 12 cm.** **Why:** Area: 3 + 2 + 2 = 7 cm². Perimeter: the figure fits in a 3 by 3 box and has no dents on the outside, so 2 × (3 + 3) = 12 cm.
4 page 12	**Answer: 7** **Why:** Length + width is 22 ÷ 2 = 11. The length is 11 - 4 = 7 cm.